POWER BI DATA ANALYTICS PROJECT

Blinkit
Sales & Outlet Performance Analysis

A descriptive business intelligence project based on a synthetic Blinkit-style dataset

Prepared as: Academic Project Report

Tools: Microsoft Excel • Power BI Desktop • Power Query • DAX

Dataset: 8,523 generated records | 12 fields

Data integrity note: This workbook is synthetic and created for academic practice. It is not an official Blinkit dataset and does not represent verified company performance.

2. Executive Summary

This report demonstrates a retail sales analysis workflow aligned with the uploaded Blinkit Python notebook. The notebook's analytical sequence includes data import, dataset inspection, fat-content normalization, KPI calculation and visual analysis by fat content, item type, outlet tier, establishment year, outlet size and outlet location.

The practice workbook contains 8,523 generated records and 12 fields. It supports the notebook's main chart requirements and adds a data dictionary and KPI summary.

KPI	Synthetic sample result
Total Sales	■2,241,202.31
Average Sales per record	■262.96
Record count	8,523
Average rating	3.95/5
Average item MRP	■147.04
Average item visibility	0.099

The notebook labels Sales sum as both Total Sales and Number of Items Sold. These are not the same: the dataset has no quantity field, so this report uses record count and does not claim units sold.

3. Introduction

Blinkit is an Indian quick-commerce platform focused on delivering groceries and everyday essentials. Retail performance can be studied through product attributes, sales, outlet characteristics and customer ratings. A structured dataset allows analysts to compare categories and outlet segments rather than reviewing each record separately.

This project converts the workflow in the uploaded notebook into a Power BI-oriented report. It uses descriptive analytics to summarize the supplied field structure as reflected in the notebook, while generating a synthetic dataset because the notebook refers to a CSV file that was not uploaded.

- Understand product and outlet sales patterns.
- Compare sales by item fat content and item type.
- Study sales across outlet location tiers and sizes.
- Explore whether outlet establishment year is associated with sales totals.
- Present clear KPIs and visuals for business discussion.

4. Project Objectives & Scope

Objectives

• Create a Power BI-ready Blinkit-style practice dataset.

• Replicate the notebook's data cleaning of Item Fat Content values.

• Calculate total sales, average sales and average rating correctly.

• Visualize sales by fat content, item type, outlet tier, establishment year, size and location.

• Develop a dashboard structure and practical recommendations.

Scope and limitation

The synthetic dataset contains 8,523 records and 12 fields. It has no order date, order ID, quantity, customer ID, delivery time, stock quantity, discount, profit or return reason. Consequently, it supports descriptive sales analysis but not customer cohort analysis, time-series sales trends, units sold, margin analysis or delivery performance.

5. Business Context

Quick-commerce businesses balance product assortment, inventory availability, pricing and local fulfilment. Item type indicates the product category, while fat content is a product attribute relevant to selected grocery categories. Outlet location tier, size and establishment year provide store-level dimensions for comparing sales.

Dimension	Business question
Item type	Which categories contribute more sales?
Fat content	How do Low Fat and Regular products compare?
Outlet tier	How does sales contribution differ by location tier?
Outlet size	Which outlet-size segment contributes more?
Establishment year	How do outlet cohorts compare in total sales?
Rating	What is the average customer rating in the sample?

The data does not establish causation. Differences across outlet groups may reflect sample design, product mix or other factors not captured in the workbook.

6. Dataset Description

The workbook has 8,523 rows and 12 columns. The fields follow the 12-column retail dataset structure used in the uploaded notebook.

Field	Description
Item Identifier	Generated product/item code
Item Weight	Item weight in kg
Item Fat Content	Low Fat or Regular
Item Visibility	Simulated shelf-visibility proportion
Item Type	Product category
Item MRP	Maximum retail price in INR
Outlet Identifier	Generated outlet code
Outlet Establishment Year	Year outlet was established
Outlet Size	Small, Medium or High
Outlet Location Type	Tier 1, Tier 2 or Tier 3
Sales	Simulated sales value in INR
Rating	Simulated customer rating (1–5)

7. Dataset Preview

The image below shows the first 12 rows and selected fields from the generated workbook. Open Blinkit_Data in Excel to view all records.

Synthetic Blinkit-Style Dataset — First 12 Records

Item Identifier	Item Fat Content	Item Type	Outlet Location Type	Outlet Establishment Year	Outlet Size	Sales	Rating
IT043368	Regular	Frozen Foods	Tier 3	1998	High	354.31	3.4
IT019215	Low Fat	Fruits and Vegetables	Tier 3	2007	Small	194.84	4.7
IT067365	Regular	Canned	Tier 2	2002	Small	273.68	3.7
IT065988	Low Fat	Canned	Tier 3	2009	Medium	248.31	4.3
IT073929	Low Fat	Baking Goods	Tier 1	1985	Small	470.08	3.3
IT044411	Regular	Frozen Foods	Tier 1	2002	High	177.74	4.1
IT104681	Low Fat	Meat	Tier 2	2009	High	230.87	4.1
IT099568	Low Fat	Soft Drinks	Tier 3	1999	Small	178.53	3.6
IT012027	Regular	Fruits and Vegetables	Tier 3	2004	Medium	239.98	5.0
IT074425	Regular	Baking Goods	Tier 1	2007	High	81.09	4.2
IT032144	Low Fat	Household	Tier 1	2004	Small	550.62	4.0
IT049818	Regular	Frozen Foods	Tier 2	2007	Medium	284.36	4.0

The uploaded notebook references /content/blinkit_data.csv, but that CSV was not included with the notebook. Therefore, the workbook provided with this report is a clearly labelled synthetic replacement for practice.

8. Tools & Technologies

Tool	Role
Microsoft Excel	Stores the generated data, field descriptions and KPI summary.
Power BI Desktop	Creates interactive report pages, slicers, KPI cards and charts.
Power Query	Imports data, checks data types, normalizes categories and handles quality checks.
DAX	Defines total sales, average sales, record count and average rating measures.
Python / Pandas	Generates the synthetic dataset and analysis charts for this report.
Matplotlib / Seaborn	Creates report illustrations for category, outlet and heatmap analysis.

9. Methodology & Workflow

- Review the uploaded notebook to identify its fields, transformations, KPIs and chart requirements.
- Generate a synthetic dataset with the same 12-field structure.
- Inspect row count, column names, data types, unique values and sample records.
- Normalize Item Fat Content values: LF and low fat become Low Fat; reg becomes Regular.
- Calculate total sales, average sales and average rating.
- Build category and outlet visuals that match the notebook's analysis.
- Import the Excel workbook into Power BI and recreate the visuals with measures and slicers.
- Interpret the descriptive patterns and document limitations.

10. Data Cleaning & Validation

Notebook step	Power BI / Power Query implementation
Inspect unique fat-content labels	Use Column distribution or distinct values on Item Fat Content.
Normalize abbreviations	Replace LF and low fat with Low Fat; replace reg with Regular.
Set data types	Text for categories/IDs; decimal for weight, visibility, MRP, sales and rating; whole number for year.
Check missing values	Inspect null counts for every field; investigate before replacing.
Check ranges	Rating should be 1–5; visibility should be non-negative; MRP and Sales should be positive.
Check identifiers	Review Item Identifier and Outlet Identifier formats; do not remove repeated product codes without reason.

The generated dataset has normalized fat-content labels. A real source CSV should be retained separately so that cleaning steps can be audited.

11. Data Model & DAX Measures

For this dataset, a single fact-style table is sufficient for a beginner dashboard. If the project grows, create separate Item and Outlet dimension tables and connect them to the sales table using stable keys.

```
Total Sales = SUM(Blinkit_Data[Sales])

Average Sales = AVERAGE(Blinkit_Data[Sales])

Record Count = COUNTROWS(Blinkit_Data)

Average Rating = AVERAGE(Blinkit_Data[Rating])

Average Item MRP = AVERAGE(Blinkit_Data[Item MRP])

Average Visibility = AVERAGE(Blinkit_Data[Item Visibility])
```

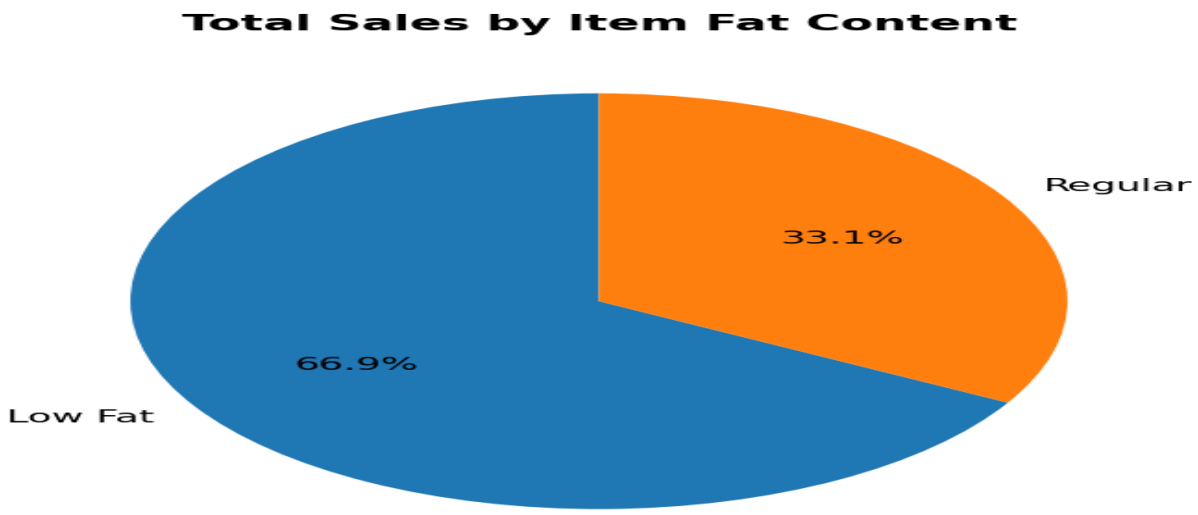
Important: Record Count is not the same as units sold. The source notebook does not include an item quantity field, so a units-sold measure cannot be calculated from the documented schema.

12. KPI Analysis

KPI	Value	Meaning
Total Sales	■2,241,202.31	Sum of Sales across all records.
Average Sales	■262.96	Mean Sales value per record.
Record Count	8,523	Number of dataset rows.
Average Rating	3.95/5	Mean of Rating values.
Average MRP	■147.04	Mean listed item MRP.
Average Visibility	0.099	Mean Item Visibility value.

Use KPI cards at the top of the Power BI overview page. Ensure each card responds to slicers so users can compare selected item types or outlet tiers.

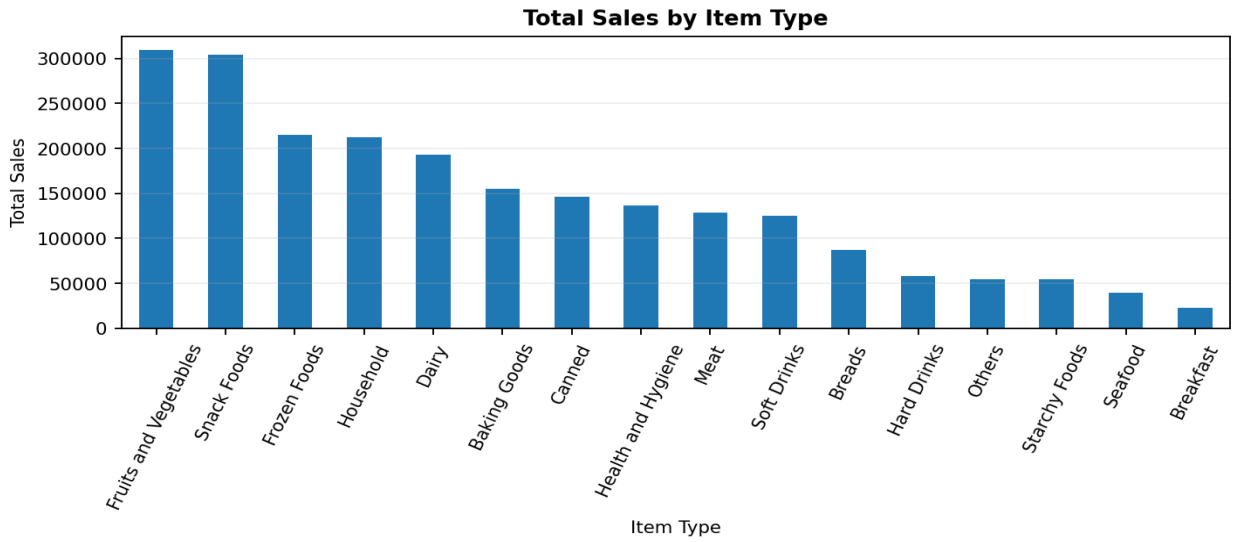
13. Total Sales by Fat Content



This chart compares the sales contribution of Low Fat and Regular items. It follows the first chart requirement in the notebook. The sample's proportions are illustrative because records were generated for this project.

In Power BI, use a donut or pie chart with Item Fat Content as Legend and Total Sales as Values.

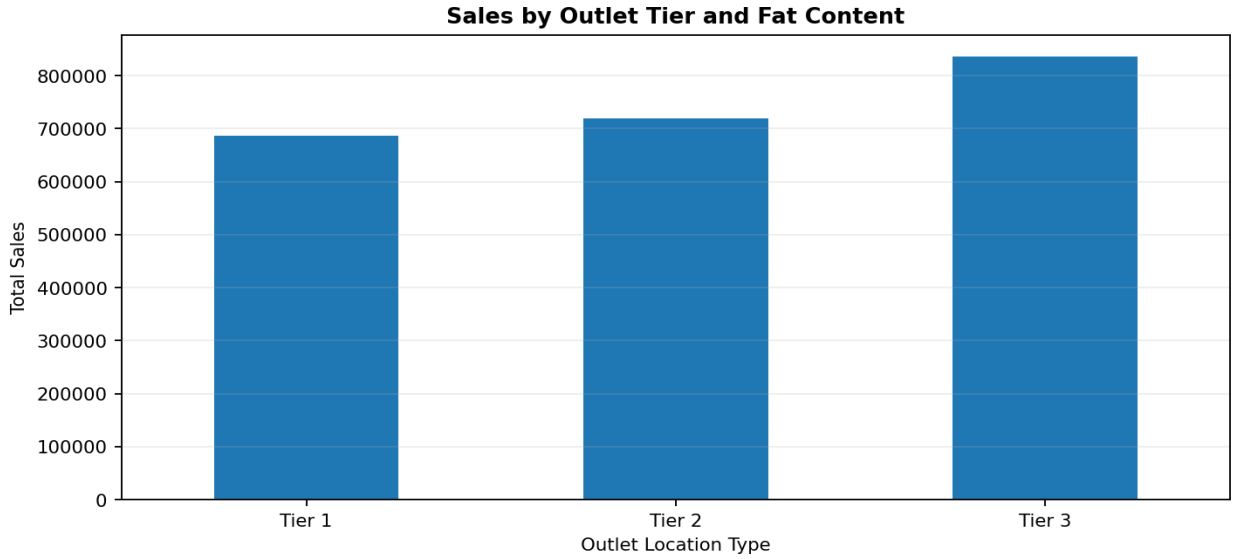
14. Total Sales by Item Type



The item-type ranking identifies categories with higher sales totals in the sample. A descending bar chart is useful because it makes category comparison easier than an unsorted list.

Management interpretation should consider assortment, availability, local demand and category-level margins. Margin and inventory data are not included here.

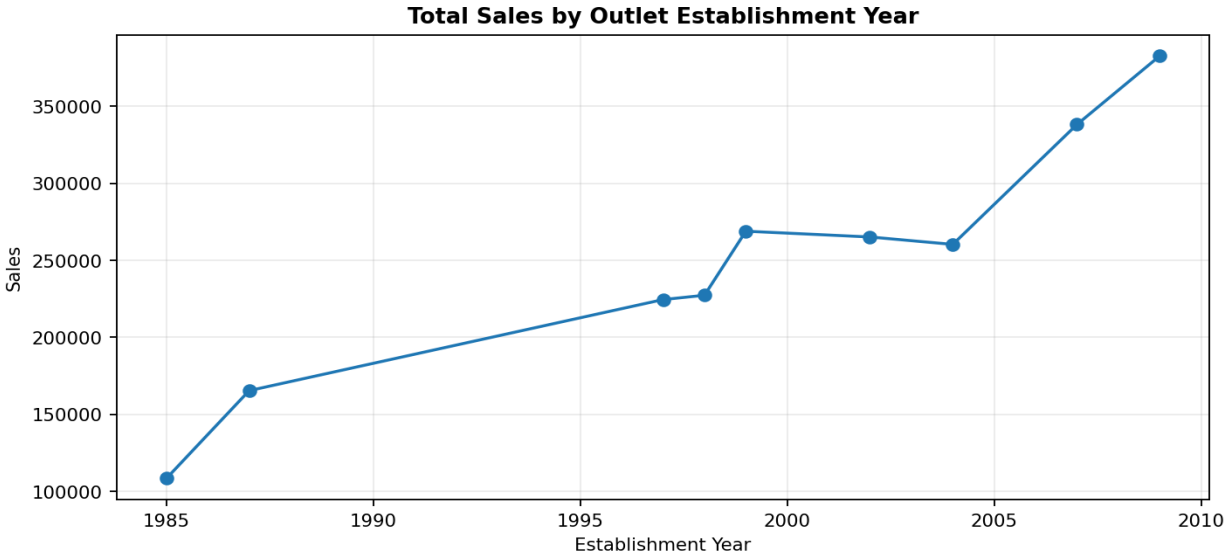
15. Fat Content by Outlet Tier



This view compares the total sales of fat-content groups across Tier 1, Tier 2 and Tier 3 outlet locations. It follows the notebook's grouped bar chart requirement.

In Power BI, use a clustered column chart: Axis = Outlet Location Type; Legend = Item Fat Content; Values = Total Sales.

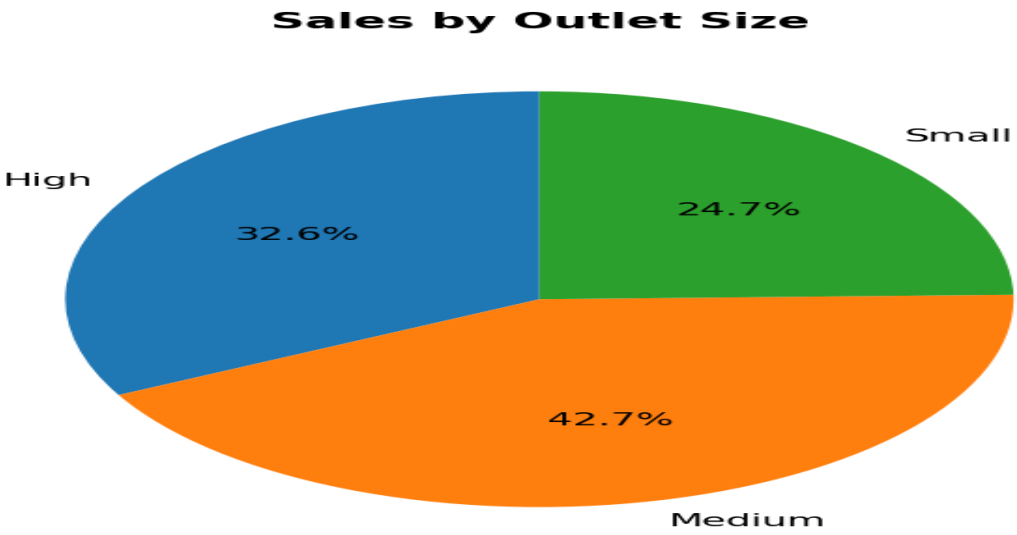
16. Sales by Outlet Establishment Year



The line chart compares sales totals by outlet establishment year. It is a cohort-style comparison, not a time series of sales over calendar dates. The dataset has no transaction date.

A high total for an establishment-year group does not prove that newer or older outlets are more productive; outlet counts and operating duration would be needed for a fair comparison.

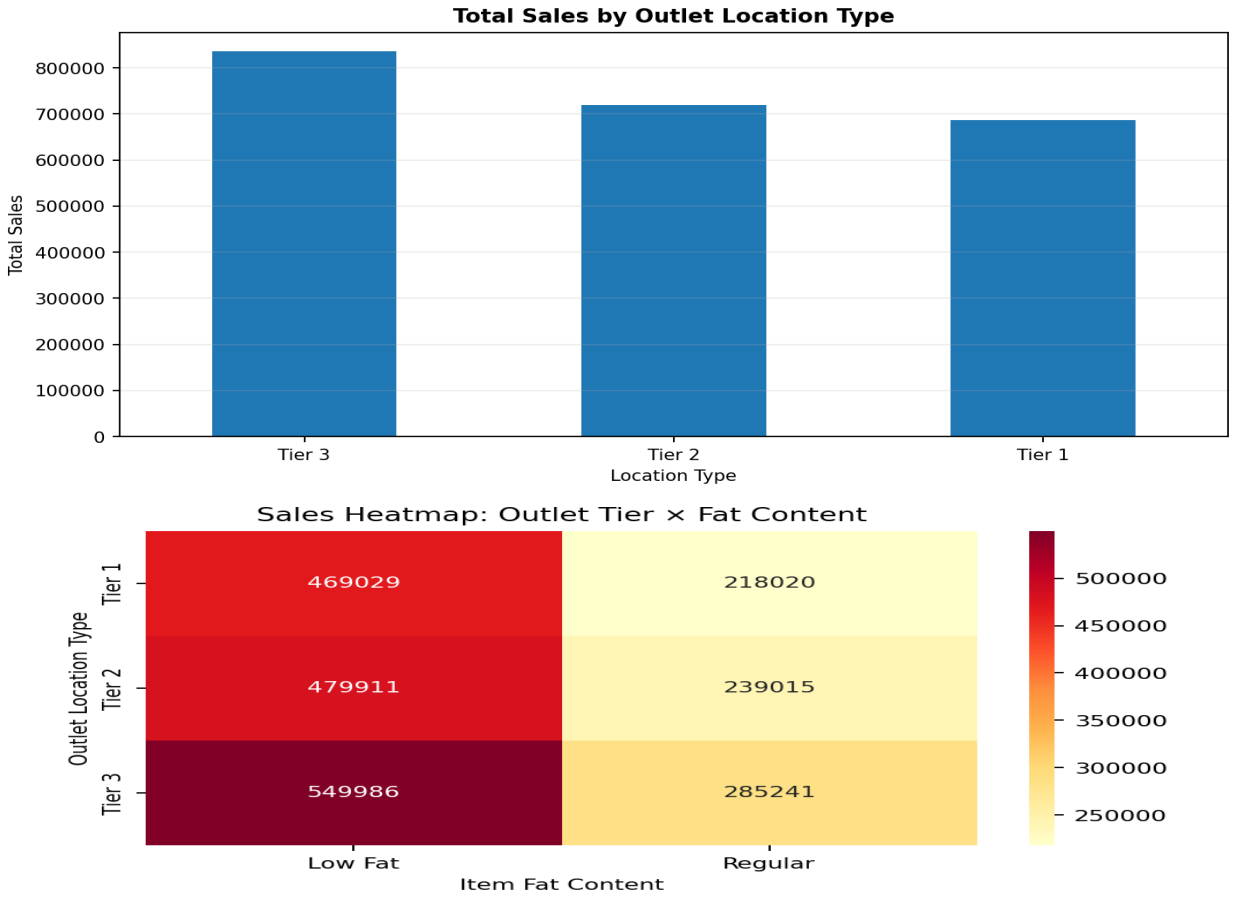
17. Sales by Outlet Size



The pie chart displays each outlet-size segment's share of total sales. In the notebook, the field is named Outlet Size and the categories are grouped into size bands.

For a management dashboard, also show absolute sales and the number of outlets in each size segment. Sales share alone can be influenced by the number of outlets in each group.

18. Sales by Location Tier & Heatmap



The bar chart ranks Tier 1, Tier 2 and Tier 3 by total sales. The heatmap combines outlet tier with fat content to make the two-dimensional comparison easier.

19. Results, Insights & Recommendations

Findings supported by this sample

• Sales can be segmented by product fat content and item type.

• Outlet tier and outlet size provide useful comparison dimensions.

• Establishment year can be used to group outlets into cohorts.

• Average rating adds a simple customer-experience indicator.

• A normalized category field improves consistency in filtering and chart legends.

Actionable recommendations to validate with real data

• Review high-sales item categories for assortment and replenishment planning.

• Compare outlet tiers using both total sales and outlet counts to avoid misleading scale effects.

• Investigate rating differences by product and outlet before taking corrective action.

• Add inventory, quantity, order date, discount and cost fields in a future dataset to enable deeper analysis.

• Validate all patterns with authorized operational data before making business decisions.

20. References & Appendix

Source and references

• Uploaded notebook: Puja_Sah (1).ipynb — source for the Blinkit analysis structure, column names, fat-content cleaning logic, KPI intent and chart requirements.

• Microsoft Learn — Power BI documentation: https://learn.microsoft.com/power-bi/

• Microsoft Learn — Power Query documentation: https://learn.microsoft.com/power-query/

• Microsoft Learn — DAX documentation: https://learn.microsoft.com/dax/

• Blinkit website: https://blinkit.com/

Appendix A — Workbook sheets

• Blinkit_Data: 8,523 synthetic records and 12 fields.

• Data_Dictionary: definitions and data types.

• KPI_Summary: measures, values and calculation descriptions.

Appendix B — Suggested Power BI pages

• Overview: Total Sales, Average Sales, Record Count and Average Rating cards.

• Product Analysis: sales by fat content and item type.

• Outlet Analysis: fat content by tier, establishment year and outlet size.

• Location Analysis: tier ranking and tier × fat-content heatmap.

All figures are calculated from the generated practice dataset, not actual Blinkit business records.